Choose a motion model before an equation

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Constant acceleration / ES: aceleración constante / PT: aceleração constante

Velocity changes by the same amount each second.

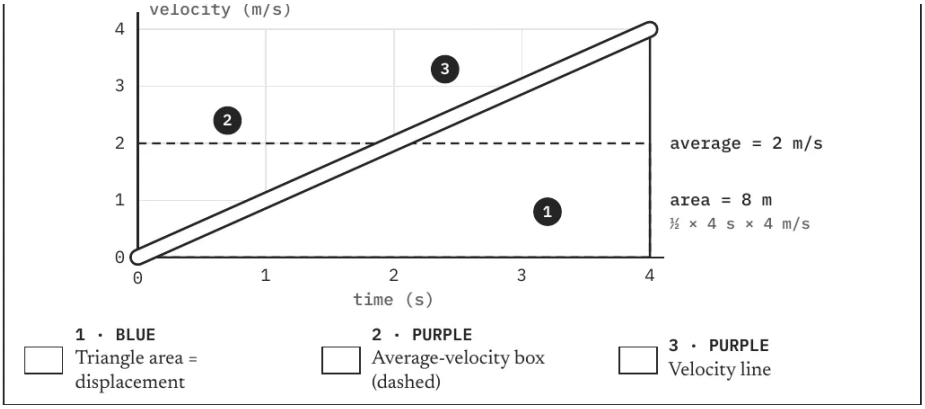
Also on your note page: Steady velocity, Average velocity, Model selection.

B READ THE RELATIONSHIP

Constant acceleration means equal changes in velocity in equal times.

Choose the equation by the given and missing quantities

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart speeds up steadily from 0 to 8 m/s in 4 s. Find its displacement.

Given $v_i = 0$, $v_f = 8$ m/s $t = 4$ s constant a

Rule $\Delta x = \frac{1}{2}(v_i + v_f) \times t$

Work average $= (0 + 8) \div 2 = 4$ m/s $\Delta x = 4 \times 4$

Answer +16 m average works: a is constant

E SAVE FOR THE FINAL MEETING

Steady velocity or steady acceleration? Choosing the right model for XJ is the difference between a forecast that holds and one that drifts.

Read the physics: Constant acceleration

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

List what you know and what you want. A steady velocity needs no acceleration. A steady change in velocity needs a constant-acceleration model.

02 HOW IT WORKS

On a velocity–time graph, slope gives acceleration. The signed area gives displacement. For constant acceleration, the average of the first and last velocities gives the average velocity.

03 WHAT TO CHECK

Choose an equation that uses your known values. Then check the sign and the unit. A matching graph gives a second way to test the result.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart speeds up steadily from 0 to 8 m/s in 4 s. Find its displacement. +16 m average works: a is constant.

05 CONNECT TO XJ

Steady velocity or steady acceleration? Choosing the right model for XJ is the difference between a forecast that holds and one that drifts.

SAY IT BEFORE YOU WRITE

What must stay constant to average the endpoint velocities? I chose ___ because I know ___ and need ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Steady velocity

ES: velocidad constante / PT: velocidade constante

Meaning in my words	How can you tell?
How would you explain steady velocity to a friend? In my words, this means ___.	Photographs are one second apart. What pattern would you look for? I can tell because ___.
Example or sketch	**Non-example and why**
Draw three positions for a cart moving 2 m right each second. My example shows ___.	A bike circles a track at a steady speed. Is its velocity constant? I would change ___ because ___.

Constant acceleration

ES: aceleración constante / PT: aceleração constante

Meaning in my words	How can you tell?
How would you explain constant acceleration to a friend? In my words, this means ___.	The velocities are 1, 3 and 5 m/s one second apart. What repeats? I can tell because ___.
Example or sketch	**Non-example and why**
If those gains continue, what velocity would you write for the next second? My example shows ___.	Someone adds 2 m to position every second and calls that speeding up. What pattern did they draw? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Average velocity

ES: velocidad media / PT: velocidade vetorial média

Meaning in my words	How can you tell?
How would you explain average velocity to a friend? In my words, this means ___.	A cart travels out and back. Which two points determine its average velocity? I can tell because ___.
Example or sketch	**Non-example and why**
Right is positive. A cart goes from +8 m to +2 m in 3 s. What is its average velocity? My example shows ___.	"Average the two speeds, and you always get average velocity." What is missing? I would change ___ because ___.

Model selection

ES: selección de modelo / PT: escolha do modelo

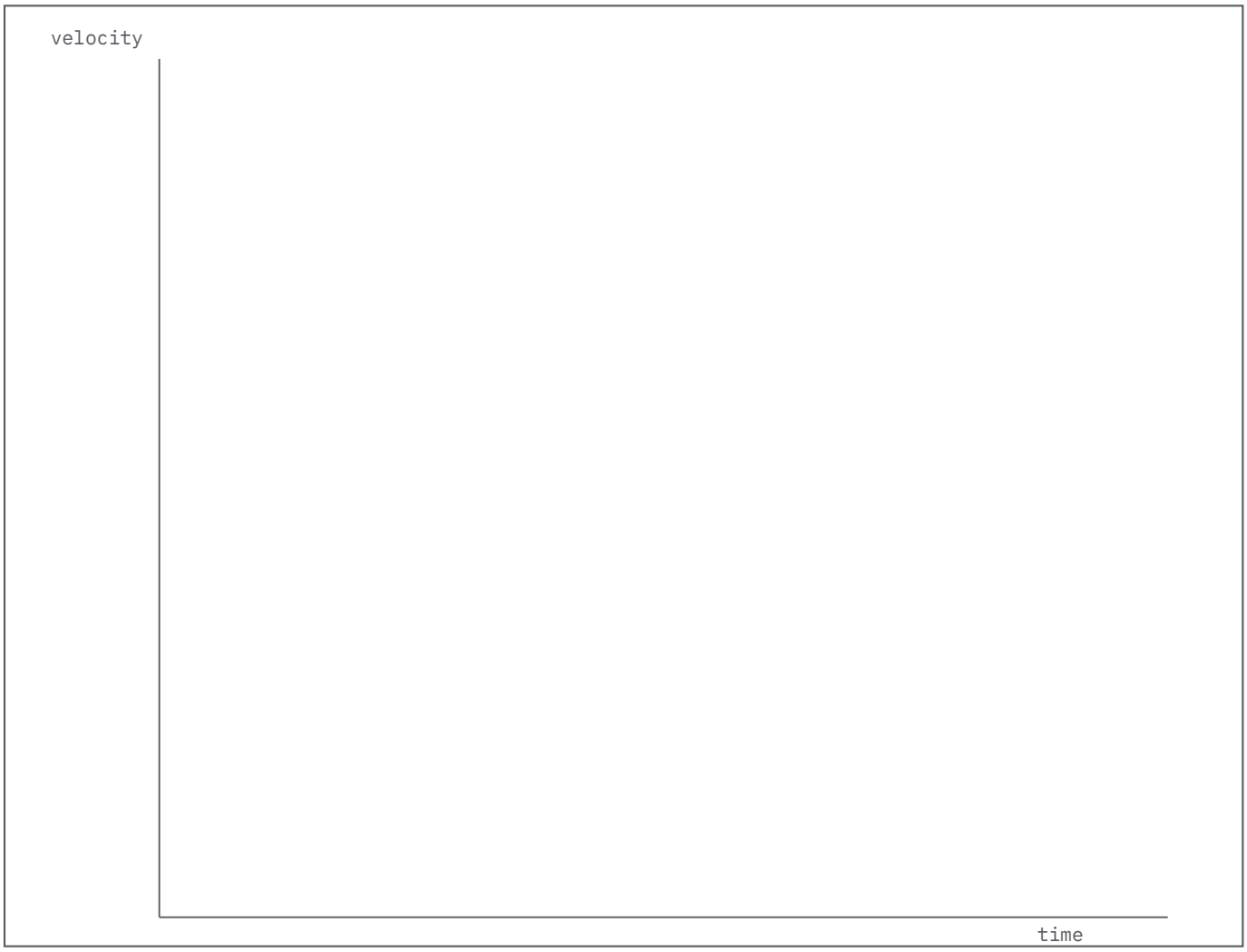
Meaning in my words	How can you tell?
How would you explain model selection to a friend? In my words, this means ___.	Which model fits velocity readings of 2, 4 and 6 m/s one second apart? I can tell because ___.
Example or sketch	**Non-example and why**
Which model would you use for a cart moving straight at 3 m/s every second? My example shows ___.	A friend chooses a formula because it is the shortest. What should you ask first? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch a straight velocity–time line. Label which feature gives acceleration and which gives displacement.

Labels you can use: given · wanted · constant.



Tell a partner

I chose ___ because I know ___ and need ___.

Catch the next shot

08

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use a velocity-time graph to predict displacement.

CHOOSE ONE JOB TOGETHER

CAMERA ENGINEER ☐

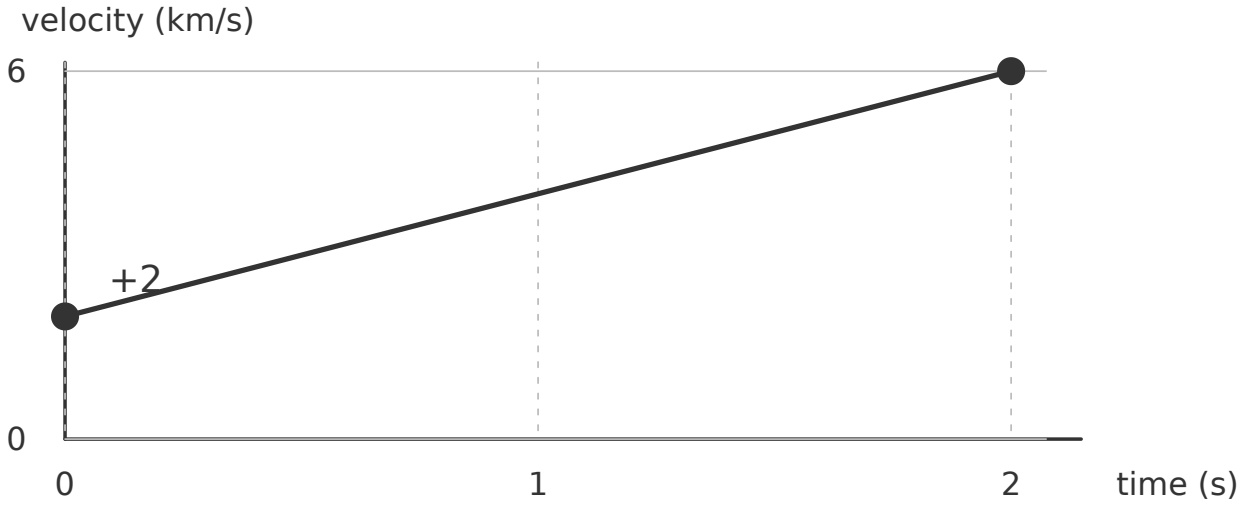
Shade the area under the velocity-time trace. Send the camera's displacement at the end of the shot.

REPLAY EDITOR ☐

Explain why using the final speed for the whole clip would miss the shot. Point to the graph as you speak.

Word help: area / constant / displacement

XJ moves from **+2** to **+6 km/s** over **2 s**. The rehearsal assumes constant acceleration. Where should the next camera look relative to the start?



01 OUR FIRST CALL

Where should the camera look relative to the start? Show how the graph helps.

Our forecast is ___ if ___ stays constant.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Does the new report still let you guarantee that displacement? Explain your revised call.

Now we would say ___ because ___.

Switch speakers. What does your shaded area measure?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks